

WITTMUNDHAFEN, Germany

WMO: 101260

Lat: 53.5503N

Lon: 7.6672E

Elev: 12

StdP: 101.18

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.8	-5.8	-10.5	1.5	-6.4	-8.3	1.9	-4.9	14.6	7.4	13.1	7.5	3.8	100	0.590

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.1	29.1	19.8	26.8	19.1	24.4	18.0	20.8	27.2	19.7	25.1	18.7	23.4	4.2	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	13.5	23.0	17.7	12.7	22.2	16.9	12.0	21.3	59.8	27.5	56.1	25.1	53.0	23.5	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.3	9.7	8.5	-10.4	32.9	3.0	1.4	-12.6	33.9	-14.3	34.7	-16.0	35.5	-18.2	36.5		
(5)			-10.6	22.9	2.8	0.8	-12.6	23.5	-14.3	24.0	-15.8	24.4	-17.9	25.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	2.3	2.8	5.1	9.0	12.5	15.4	18.0	17.9	14.7	10.6	6.2	3.3	(6)
(7)		DBStd	6.59	4.17	3.89	3.37	3.61	3.46	3.14	3.24	2.91	2.63	3.23	3.35	4.13	(7)
(8)		HDD10.0	1033	238	201	155	62	13	0	0	0	1	32	120	210	(8)
(9)		HDD18.3	3188	497	434	411	282	186	101	47	44	115	241	364	468	(9)
(10)		CDD10.0	974	0	1	3	31	89	161	249	244	140	49	6	1	(10)
(11)		CDD18.3	86	0	0	0	0	3	12	37	29	5	0	0	0	(11)
(12)		CDH23.3	731	0	0	0	6	33	114	309	232	36	1	0	0	(12)
(13)		CDH26.7	209	0	0	0	0	4	31	97	73	4	0	0	0	(13)
(14)	Wind	WSAvg	4.1	4.9	4.7	4.7	4.1	3.9	3.7	3.5	3.3	3.5	3.9	4.1	4.5	(14)
(15)	Precipitation	PrecAvg	825	65	53	45	42	54	69	91	86	78	77	75	(15)	
(16)		PrecMax	1144	139	131	98	118	117	154	159	188	220	205	172	197	(16)
(17)		PrecMin	540	1	21	6	3	10	8	38	26	18	17	4	20	(17)
(18)		PrecStd	131	36	26	27	25	24	35	36	36	48	39	39	43	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.9	13.0	17.0	23.9	26.9	30.3	32.3	31.8	27.0	21.1	15.2	12.8	(19)
(20)			MCWB	10.2	9.5	10.8	14.8	17.8	20.5	20.9	20.3	19.6	16.5	13.2	11.3	(20)
(21)		2%	DB	10.1	10.7	13.9	20.5	23.8	26.8	29.4	28.8	23.9	18.1	13.6	11.2	(21)
(22)			MCWB	8.8	8.6	9.8	13.0	15.9	18.6	20.1	19.7	18.0	15.1	12.0	9.9	(22)
(23)		5%	DB	8.9	9.3	11.8	17.9	21.2	23.7	26.8	25.8	21.1	16.4	12.1	9.9	(23)
(24)			MCWB	7.8	7.8	8.8	12.3	15.0	17.5	19.4	18.9	16.4	14.0	10.6	8.9	(24)
(25)		10%	DB	7.8	8.1	10.1	15.2	19.0	21.2	24.2	23.3	19.1	15.1	10.9	8.7	(25)
(26)			MCWB	6.8	6.7	8.0	10.9	14.0	16.1	18.4	17.8	15.4	13.2	9.5	7.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.5	10.4	12.2	15.6	19.3	21.2	22.3	22.0	20.1	16.9	13.7	11.6	(27)
(28)			MCDWB	11.6	11.7	15.0	21.8	24.7	28.6	29.4	28.3	25.6	20.3	15.0	12.5	(28)
(29)		2%	WB	9.0	8.9	10.5	13.9	17.0	19.5	20.9	20.5	18.4	15.4	12.3	10.2	(29)
(30)			MCDWB	9.8	10.1	12.8	19.1	22.1	25.2	27.7	26.2	23.2	17.5	13.5	11.2	(30)
(31)		5%	WB	7.8	7.9	9.4	12.6	15.5	18.0	19.9	19.3	17.1	14.4	10.9	8.9	(31)
(32)			MCDWB	8.6	9.2	11.5	17.0	19.9	22.4	25.4	24.5	20.2	16.1	12.1	9.7	(32)
(33)		10%	WB	6.7	6.7	8.1	11.3	14.2	16.7	18.8	18.4	16.0	13.5	9.7	7.6	(33)
(34)			MCDWB	7.5	7.9	9.9	15.0	18.3	20.4	23.4	22.8	18.5	15.0	10.6	8.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.5	5.4	7.0	9.0	9.1	9.0	9.1	9.2	8.1	6.9	5.1	4.5	(35)
(36)			MCDBR	5.0	6.7	10.3	13.3	13.0	13.4	13.4	13.1	11.3	8.3	5.7	4.8	(36)
(37)			MCWBR	4.5	5.1	6.8	7.3	6.9	7.0	6.0	5.9	6.2	5.6	4.7	4.5	(37)
(38)		5% WB	MCDWB	4.7	5.7	9.1	11.8	11.6	11.9	12.0	11.4	9.9	7.6	5.4	4.7	(38)
(39)			MCWBR	4.5	4.8	6.4	7.1	6.7	6.8	5.9	5.8	5.9	5.4	4.8	4.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.328	0.346	0.375	0.407	0.402	0.401	0.415	0.410	0.387	0.362	0.341	0.320	(40)	
(41)		taud	2.314	2.329	2.272	2.238	2.292	2.320	2.295	2.322	2.375	2.402	2.347	2.305	(41)	
(42)		Ebn at Noon	629	744	798	820	846	850	828	810	781	718	606	558	(42)	
(43)		Edh at Noon	60	81	104	123	122	120	121	112	94	75	58	51	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.57	1.22	2.46	4.10	5.12	5.36	5.08	4.22	2.99	1.66	0.70	0.43	(44)	
(45)		RadStd	0.06	0.18	0.26	0.41	0.44	0.40	0.60	0.33	0.30	0.17	0.07	0.05	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.47	+1.38	+1.54	N/A	N/A	N/A	N/A	-225	+62	N/A
(47) Regional (48 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	-131	N/A	N/A

Nomenclature: See separate page